

EDUCATION

B.Sc. Computer and Information Sciences, Suez Canal University
AWS Certified Solutions Architect – Associate, Amazon Web Services

2022 – 2026
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EXPERIENCE

Backend & Infrastructure Engineer
Repovive

Nov 2025 – Present
Part Time – United States, Remote

A competitive-programming platform pairing scalable automated judging infrastructure and contest tooling with AI-assisted problem authoring.

- Architected the judge infrastructure (judge1 on **Judge0**) for sandboxed, multi-language auto-grading at contest scale.
- Scaled concurrent execution **10x (50 to 500+ processes)** with an async, containerized job-orchestration pipeline.
- Cut cloud infrastructure cost **95% (\$600 to \$25/mo)** by migrating serverless workloads to a Docker/Nginx/Redis stack.
- Sandboxed **20+ languages** with resource- and syscall-limited Judge0 Docker images for secure multi-tenant contests.
- Built a real-time collaborative code workspace using **WebSockets** and **Yjs CRDTs** over a Monaco editor.
- Co-built **Build v2**, an AI problem-authoring IDE with a 28-tool agent pipeline guarded by server-side execution locks.
- Tech: TypeScript, Next.js, AWS, Docker, Redis, MongoDB, Nginx, Yjs.

OPEN SOURCE CONTRIBUTIONS

- GitLab Go SDK (client-go)**: typed group integrations – Slack, Jira, MS Teams, Mattermost, Harbor, +3 | **!2692, !2691, !2675, !2670.**
- GitLab Terraform provider**: new Harbor & Mattermost integration resources + group SAML attribute | **!2881, !2909, !2852.**
- GitLab monorepo**: Json.safe_parse hardening and Vue lint compliance | **!221163, !221158.**
- LLVM / Clang**: constexpr X86 cvtpd2ps eval and AArch64/ARM CLS → ISD: :CTLS lowering | **#169980, #178885, #178430.**
- Apple Swift NIO**: idempotent NIOFileSystem directory creation | **#3410.**

PROJECTS

Competitive-Programming Judge — AWS Reference Architecture

github.com/HamzaHassanain/CP-Judge-AWS

- Designed an AWS reference architecture: Judge0 isolate on ECS-on-EC2, Fargate orchestrator, Spot ASG on queue depth.
- Automated post-round cheating detection with Step Functions, AWS Batch, Lambda/Bedrock, and Athena.

Autonomous AI Tutor & Curriculum Engine, Graduation Project

github.com/sigma-loop

- Led backend & DevOps for an 8-person team building an on-demand AI course, lesson, and challenge generator.
- Graded code in a Judge0 sandbox and math proofs via a confidence-scored LLM; owned all CI/CD and AWS deployments.

Polyman CLI

npmjs.com/package/polyman-cli | github.com/HamzaHassanain/polyman

- Built a TypeScript CLI automating the Codeforces Polygon workflow: test generation, validation, and direct API sync.

Systems in C++ — HALlocator & cppress

HALlocator | cppress

- Built a Red-Black-Tree best-fit memory allocator (**O(log n)**) and an epoll-driven C++ web framework, both from scratch.

TECHNICAL WRITING

- Authored **"What Every Programmer Should Know About Memory"**, a series on CPU caches, virtual memory, and optimization.
- Published **"I Built a Backend Web Framework from Scratch in C++"**, covering non-blocking I/O, HTTP parsing, and thread pools.

ACHIEVEMENTS

ICPC ECPC & ACPC — Competitive Programming

icpc.global/YC7Z02WQCPIB

17th of 500 teams at ECPC (2024); two-time ACPC regional qualifier (2024, 2025), top 250 in the Africa & Arab region.

SKILLS

Cloud & DevOps	AWS (Solutions Architect – Associate), Docker, Nginx, Redis, Terraform, CI/CD (GitHub Actions), DigitalOcean, Linux
Backend & Full-Stack	TypeScript, Node.js, Express.js, Next.js, React, MongoDB, PostgreSQL, REST APIs, Microservices, WebSockets, CRDTs (Yjs)
Languages & Systems	C++, Go, Rust, Python, SQL; System Design, Distributed Systems, Message Queues